LDOS modulation $\delta\rho$ from the Born T-matrix sum $S(\mathbf{q})$

1 Conventions

The lattice has N sites $\mathbf{r}$ and N momenta $\mathbf{k}$ in the Brillouin zone. Operators are related by

$$\Psi_{\mathbf{r}} = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}}\, \Psi_{\mathbf{k}}, \qquad \Psi_{\mathbf{k}} = \begin{pmatrix} c_{\mathbf{k}\uparrow} \\ c^{\dagger}_{-\mathbf{k}\downarrow} \end{pmatrix}, \tag{1}$$

so that the two-point Green's function transforms as

$$G(\mathbf{r},\mathbf{r}') = \frac{1}{N} \sum_{\mathbf{k},\mathbf{k}'} e^{i\mathbf{k}\cdot\mathbf{r}}\, G(\mathbf{k},\mathbf{k}')\, e^{-i\mathbf{k}'\cdot\mathbf{r}'}. \tag{2}$$

The bare retarded and advanced Green's functions are

$$G_0^R(\mathbf{k},\omega) = [\omega + i\gamma - H_{\mathrm{BdG}}(\mathbf{k})]^{-1}, \qquad G_0^A(\mathbf{k},\omega) = \left[G_0^R(\mathbf{k},\omega)\right]^{\dagger}, \tag{3}$$

and in general $G^A(\mathbf{r},\mathbf{r}') = [G^R(\mathbf{r}',\mathbf{r})]^{\dagger}$, so on the diagonal $G^A(\mathbf{r},\mathbf{r}) = [G^R(\mathbf{r},\mathbf{r})]^{\dagger}$.

2 Born approximation and the quantity $S(\mathbf{q})$

For an impurity $H_{\mathrm{imp}} = \sum_{\mathbf{k}\mathbf{k}'} \Psi^{\dagger}_{\mathbf{k}} T(\mathbf{k},\mathbf{k}') \Psi_{\mathbf{k}'}$ with $T(\mathbf{k},\mathbf{k}')^{\dagger} = T(\mathbf{k}',\mathbf{k})$ (Hermitian), the first-order correction is

$$\delta G^R(\mathbf{k},\mathbf{k}') = G_0^R(\mathbf{k})\, T(\mathbf{k},\mathbf{k}')\, G_0^R(\mathbf{k}'). \tag{4}$$

The code computes (`g0k`, `T`, `g0kq`)

$$S(\mathbf{q}) = \sum_{\mathbf{k}} G_0^R(\mathbf{k})\, T(\mathbf{k},\mathbf{k}+\mathbf{q})\, G_0^R(\mathbf{k}+\mathbf{q}) = \sum_{\mathbf{k}} \delta G^R(\mathbf{k},\mathbf{k}+\mathbf{q}), \tag{5}$$

a 2×2 Nambu matrix. Note that both Green's functions in S are *retarded*.

3 Real-space diagonal

Setting $\mathbf{r}' = \mathbf{r}$ in (2) and writing $\mathbf{k}' = \mathbf{k}+\mathbf{q}$:

$$\delta G^R(\mathbf{r},\mathbf{r}) = \frac{1}{N} \sum_{\mathbf{k},\mathbf{q}} e^{i\mathbf{k}\cdot\mathbf{r}}\, \delta G^R(\mathbf{k},\mathbf{k}+\mathbf{q})\, e^{-i(\mathbf{k}+\mathbf{q})\cdot\mathbf{r}} = \frac{1}{N} \sum_{\mathbf{q}} e^{-i\mathbf{q}\cdot\mathbf{r}}\, S(\mathbf{q}). \tag{6}$$

The phase is $e^{-i\mathbf{q}\cdot\mathbf{r}}$: $S(\mathbf{q})$ scatters an electron from $\mathbf{k}+\mathbf{q}$ into $\mathbf{k}$, i.e. it transfers momentum $-\mathbf{q}$.

The advanced part is the Hermitian conjugate of (6):

$$\delta G^A(\mathbf{r},\mathbf{r}) = \left[\delta G^R(\mathbf{r},\mathbf{r})\right]^\dagger = \frac{1}{N}\sum_{\mathbf{q}} e^{+i\mathbf{q}\cdot\mathbf{r}}\, S(\mathbf{q})^\dagger = \frac{1}{N}\sum_{\mathbf{q}} e^{-i\mathbf{q}\cdot\mathbf{r}}\, S(-\mathbf{q})^\dagger. \tag{7}$$

$S(-\mathbf{q})^\dagger$ is the *advanced* counterpart of $S(\mathbf{q})$ with the same momentum structure. Using $T^\dagger(\mathbf{k},\mathbf{k}') = T(\mathbf{k}',\mathbf{k})$ and shifting $\mathbf{k} \to \mathbf{k}+\mathbf{q}$,

$$S(-\mathbf{q})^\dagger = \sum_{\mathbf{k}} G_0^A(\mathbf{k}-\mathbf{q})\, T(\mathbf{k}-\mathbf{q},\mathbf{k})\, G_0^A(\mathbf{k}) = \sum_{\mathbf{k}} G_0^A(\mathbf{k})\, T(\mathbf{k},\mathbf{k}+\mathbf{q})\, G_0^A(\mathbf{k}+\mathbf{q}) = \sum_{\mathbf{k}} \delta G^A(\mathbf{k},\mathbf{k}+\mathbf{q}). \tag{8}$$

So $S(\mathbf{q}) - S(-\mathbf{q})^\dagger$ is exactly $\delta G^R - \delta G^A$ at the same $\mathbf{q}$; no retarded/advanced swap is involved.

4 LDOS

The spin-up electron LDOS is

$$\delta\rho(\mathbf{r},\omega) = -\frac{1}{\pi}\,\mathrm{Im}\;\delta G^R_{00}(\mathbf{r},\mathbf{r},\omega) = -\frac{1}{2\pi i}\left[\delta G^R(\mathbf{r},\mathbf{r}) - \delta G^A(\mathbf{r},\mathbf{r})\right]_{00}. \tag{9}$$

Substituting (6) and (7),

$$\boxed{\delta\rho(\mathbf{r}) = \frac{1}{N}\sum_{\mathbf{q}} e^{-i\mathbf{q}\cdot\mathbf{r}}\,\tilde{\rho}(\mathbf{q}), \qquad \tilde{\rho}(\mathbf{q}) = -\frac{1}{2\pi i}\left[S(\mathbf{q}) - S(-\mathbf{q})^\dagger\right]_{00} = \frac{i}{2\pi}\left[S(\mathbf{q}) - S(-\mathbf{q})^\dagger\right]_{00}.} \tag{10}$$

Comparison with $(S(\mathbf{q})-S(-\mathbf{q})^\dagger)/2\pi$. This is $\tilde{\rho}(\mathbf{q})$ without the factor i: $(S(\mathbf{q})-S(-\mathbf{q})^\dagger)_{00}/2\pi = -i\,\tilde{\rho}(\mathbf{q})$. Since $\delta\rho(\mathbf{r})$ is real, transforming it back gives $-i\,\delta\rho(\mathbf{r})$, which is purely imaginary. The magnitudes are right, but real and imaginary parts are exchanged (with a sign) in both $\mathbf{q}$- and $\mathbf{r}$-space.

5 Which sign of q?

Equation (10) defines $\tilde{\rho}$ as the coefficient of $e^{-i\mathbf{q}\cdot\mathbf{r}}$. The code (and `numpy.fft.ifft2`) uses the standard convention

$$\delta\rho(\mathbf{r}) = \frac{1}{N_q^2}\sum_{\mathbf{q}} e^{+i\mathbf{q}\cdot\mathbf{r}}\,\delta\rho(\mathbf{q}) \quad\Longleftrightarrow\quad \delta\rho(\mathbf{q}) = \sum_{\mathbf{r}} e^{-i\mathbf{q}\cdot\mathbf{r}}\,\delta\rho(\mathbf{r}), \tag{11}$$

so that

$$\delta\rho(\mathbf{q}) = \tilde{\rho}(-\mathbf{q}) = \frac{i}{2\pi}\left[S(-\mathbf{q}) - S(\mathbf{q})^\dagger\right]_{00}. \tag{12}$$

The two forms are the same physics written in opposite Fourier conventions. Both contain one retarded and one advanced term. Either is correct provided the matching inverse transform is used:

- $\tilde{\rho}(\mathbf{q})$ with $\sum_{\mathbf{q}} e^{-i\mathbf{q}\cdot\mathbf{r}}$ (`fft2`, divided by N_q^2), or
- $\delta\rho(\mathbf{q})$ of (12) with $\sum_{\mathbf{q}} e^{+i\mathbf{q}\cdot\mathbf{r}}$ (`ifft2`), as in the code.

Mixing them (e.g. $\tilde{\rho}$ with `ifft2`) gives $\delta\rho(-\mathbf{r})$. That is a C_2 rotation of the image: the irrep weights are unchanged, but the B_1, B_2 and E_1 components change sign.

Swapping $G^R \leftrightarrow G^A$ throughout would instead flip the overall sign of $\delta\rho$, since $-\frac{1}{\pi}\,\mathrm{Im}\,G^A = +\frac{1}{\pi}\,\mathrm{Im}\,G^R$. It does not change the sign of $\mathbf{q}$.

6 Consequences for the symmetry analysis

Because $\delta\rho(\mathbf{r})$ is real, $\delta\rho(-\mathbf{q}) = \delta\rho(\mathbf{q})^*$. Hence Re $\delta\rho(\mathbf{q})$ is even and Im $\delta\rho(\mathbf{q})$ is odd under $\mathbf{q} \to -\mathbf{q}$. In two dimensions inversion is the rotation $C_2 \in C_{6v}$, with $\chi(C_2) = +d_\nu$ for A_1, A_2, E_2 and $-d_\nu$ for B_1, B_2, E_1. So

$$\text{Re } \delta\rho(\mathbf{q}) \in A_1 \oplus A_2 \oplus E_2, \qquad \text{Im } \delta\rho(\mathbf{q}) \in B_1 \oplus B_2 \oplus E_1, \qquad \text{Im } \delta\rho(\mathbf{r}) = 0. \tag{13}$$

7 Implementation notes

- $S(-\mathbf{q})$ does not need a separate calculation. On the centred q-mesh, $\mathbf{q} = (m_1\mathbf{b}_1 + m_2\mathbf{b}_2)/N_q$ sits at index $(m_2 + N_q/2,\, m_1 + N_q/2)$, and $-\mathbf{q}$ at index $(N_q - i) \bmod N_q$ along each axis. Since G_0^R is periodic in $\mathbf{k}$ and V is evaluated on Wigner–Seitz-folded momenta, $S(\mathbf{q} + \mathbf{G}) = S(\mathbf{q})$, so the reindexed array is exact. This matches a direct recalculation of $S(-\mathbf{q})$ to $\sim 10^{-16}$ relative.
- The overall factor $1/N$ in (10) is omitted in the code (it only sets the scale).

8 Numerical check

On a 12×12 lattice ($N_q = N_k = 12$, $d + id$ pairing, $\omega = 1.5\,\Delta_{\text{gap}}$, $\gamma = 10^{-2}$), with a generic Hermitian $V(\mathbf{k}, \mathbf{k}')$ that has no C_{6v} symmetry:

- $\delta G_{00}^R(\mathbf{k}, \mathbf{k}')$ was built in the full $(\mathbf{k}, \mathbf{k}')$ basis, rotated to sites using $\langle \mathbf{r}|\mathbf{k}\rangle = e^{i\mathbf{k}\cdot\mathbf{r}}/\sqrt{N}$, and $-\frac{1}{\pi}\,\text{Im } \delta G_{00}^R(\mathbf{r}, \mathbf{r})$ taken directly.
- The code's path ((12) then `ifft2`) reproduces it with ratio 1 ± 10^{-12}, and its imaginary part is $\sim 10^{-16}$.
- $i\,(S(\mathbf{q}) - S(-\mathbf{q})^\dagger)_{00}/2\pi$ with the $e^{-i\mathbf{q}\cdot\mathbf{r}}$ inverse transform also reproduces it (relative error 5×10^{-15}).
- Replacing $G_0^R \to G_0^A$ in S gives exactly $-\delta\rho(\mathbf{r})$.